

Relative binning for gravitational-wave microlensing in the Chang–Refsdal model

Tejaswi Venumadhav¹

¹*Department of Physics, University of California, Santa Barbara, California 93106, USA*

(Dated: July 15, 2026)

Stars in a lensing galaxy can microlens a strongly lensed gravitational wave and imprint diffraction and interference fringes across the detector band. Relative binning accelerates gravitational-wave parameter estimation by representing nearby waveforms through smoothly varying ratios that need to be evaluated at only a sparse set of frequencies. Applied directly to a microlensed waveform, this method loses its advantage because small changes in the lens parameters shift the fringes and make the ratio rapidly varying. We reorganize the Chang–Refsdal amplification factor as a coherent sum in which the image-delay phases are kept explicit and the remaining complex amplitudes vary slowly with frequency. Away from caustics these amplitudes reduce to the individual image contributions. At moderate frequencies and through folds and cusps, we use the full wave-optics result to correct the separated contributions while preserving their sum exactly. We evaluate the total amplification from the analytic point-mass solution through source derivatives and include external convergence by an exact rescaling, without numerical integration over the lens plane. In the examples studied here, interpolating the separated amplitudes while treating the delay phases analytically reduces the required frequency sampling by more than an order of magnitude compared with direct interpolation of the full amplification ratio.

I. INTRODUCTION

Gravitational waves retain phase coherence over the frequency band of an interferometric detector. A gravitational lens can therefore produce observable diffraction and interference when the delay between competing propagation paths is comparable to the inverse wave frequency [1, 2]. Stellar-mass objects lie in this regime for ground-based detectors. Their influence is enhanced when the source is also strongly lensed by a galaxy, since the stellar population in the lens galaxy perturbs each macro image and can imprint frequency-dependent modulations on the signal [3–5]. On the scale of a single microlens Einstein radius, the smooth galaxy potential is described by a constant convergence and shear. A point mass in this local field is the Chang–Refsdal lens [6–8].

Bayesian parameter estimation requires repeated evaluations of the gravitational-wave likelihood. Relative binning accelerates this calculation by choosing a waveform near the high-likelihood region as a fiducial model, expressing each candidate as a frequency-dependent ratio to that model, and evaluating the likelihood from precomputed sums on a sparse frequency grid [9–12]. For nearby compact-binary waveforms, the amplitude and phase of this ratio vary slowly with frequency, so only a modest number of frequency samples is needed.

A wave-optics amplification factor does not share this property when it is treated as a single function. The signal is a coherent sum of contributions with different image delays. Changing the lens parameters moves the interference fringes, and the candidate-to-fiducial ratio can develop rapid oscillations and sharp features near the zeros of the fiducial amplification. This occurs even when the candidate and fiducial sources remain within the same two-image or four-image region. A change in image topology creates a second difficulty, but it is not

the origin of the failure of ordinary relative binning.

The stationary-phase expansion suggests how to reorganize the calculation. Each geometrical image contributes a phase fixed by its time delay, multiplied by a factor that changes much more slowly with frequency. We keep the delay phases separate and interpolate the accompanying factors. The isolated-image expansion alone is not accurate at moderate frequency or near a caustic, so we use it only to define the resolved-image limit. At finite frequency we assign the difference between its coherent sum and the full wave-optics result to the same separated contributions. Their sum then reproduces the full amplification exactly, while each contribution approaches the corresponding image in the high-frequency limit.

A decomposition tied directly to the physical images cannot be continued through a fold or cusp. To obtain a representation that is valid throughout the positive-parity Chang–Refsdal caustic network, we keep four labels at all parameter values. Labels without a real image are placed at the nearest critical point. When images are created, they inherit these labels continuously. An algebraic projection enforces the exact total amplification for any mixture of real and virtual labels.

The paper develops the representation in the order in which its physical motivation arises. Section II introduces the amplification factor and the analytic source-plane operator used to evaluate it. Section III relates the observed fringes to interference between geometrical images. Section IV shows why direct relative binning fails. Section V constructs separated image contributions within a region of fixed topology. Folds, cusps, and the global four-label extension are discussed in Secs. VI and VII. The corresponding multi-component relative-binning likelihood is derived in Sec. VIII. Numerical tests are presented in Sec. IX.

II. WAVE OPTICS IN THE CHANG–REFSDAL LENS

A. Fermat potential and amplification factor

The Fermat potential gives the dimensionless propagation time for a ray that crosses the lens plane at $\mathbf{x}$ when the unlensed source position is $\mathbf{y}$. Its stationary points are the geometrical images. The wave-optics amplitude is obtained by coherently summing over the complete lens plane [1, 13].

We measure angles in the Einstein-radius units of the point mass and write

$$\tau(\mathbf{x}; \mathbf{y}) = \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{y} \cdot \mathbf{x} + \frac{1}{2} |\mathbf{y}|^2 - \ln |\mathbf{x}|, \quad (1)$$

where

$$\mathbf{A} = (1 - \kappa) \mathbf{1} - \mathbf{\Gamma}, \quad (2)$$

and

$$\mathbf{\Gamma} = \gamma \begin{pmatrix} \cos 2\beta & \sin 2\beta \\ \sin 2\beta & -\cos 2\beta \end{pmatrix}. \quad (3)$$

The parameters κ , γ , and β are the external convergence, shear magnitude, and shear orientation. We consider a positive-parity macro image,

$$1 - \kappa > |\gamma|. \quad (4)$$

Macro saddle images require a different treatment of the large-scale oscillatory integral and are outside the present scope.

With the standard Fresnel normalization, the amplification factor is

$$F(w, \mathbf{y}) = \frac{w}{2\pi i} \int_{\mathbb{R}^2} d^2x \exp[iw\tau(\mathbf{x}; \mathbf{y})]. \quad (5)$$

The normalization includes the magnification of the local macro field. The dimensionless frequency is

$$w = \frac{8\pi G M_L (1 + z_L)}{c^3} f, \quad (6)$$

for a point mass M_L at redshift z_L . A lensed frequency-domain waveform is

$$\tilde{h}_L(f) = F[w(f), \mathbf{y}] \tilde{h}_U(f). \quad (7)$$

A constant added to τ changes only the overall arrival time. We subtract the minimum image delay in the figures and numerical comparisons.

Figure 1 shows representative amplification curves in the two-image and four-image regions and near a fold and cusp. The low-frequency portion is plotted on a logarithmic frequency axis. The high-frequency portion is sampled and displayed linearly because the geometrical-optics fringes are approximately periodic in w . Sampling those fringes uniformly in $\log w$ produces visual aliasing.

At low frequency the Fresnel scale samples a broad part of the Fermat surface, and the response changes slowly. At larger w , localized stationary regions interfere. The fringe spacing is governed by image-delay differences, while the contrast is governed by the relative magnifications. Several delay differences contribute inside the four-image region.

B. Evaluation from the point-mass solution

Equation (5) is the definition of the amplification, but the calculations in this paper do not evaluate it by lens-plane quadrature. The external shear is quadratic in the lens-plane coordinates and can be represented by derivatives with respect to the source position.

First set $\kappa = 0$ and remove the source phase,

$$F_{0,\gamma}(w, \mathbf{y}; \beta) = e^{iw|\mathbf{y}|^2/2} \mathcal{G}_\gamma(w, \mathbf{y}; \beta). \quad (8)$$

The exact operator identity is

$$\mathcal{G}_\gamma(w, \mathbf{y}; \beta) = \exp\left[\frac{i\gamma}{2w} D_\beta\right] \mathcal{G}_{\text{PM}}(w, \mathbf{y}), \quad (9)$$

where

$$D_\beta = \cos 2\beta (\partial_1^2 - \partial_2^2) + 2 \sin 2\beta \partial_1 \partial_2. \quad (10)$$

Writing $s = |\mathbf{y}|^2$, the point-mass seed is

$$\mathcal{G}_{\text{PM}}(w, s) = C(w) {}_1F_1\left(1 - \frac{iw}{2}; 1; -\frac{iws}{2}\right), \quad (11)$$

$$C(w) = \exp\left[\frac{\pi w}{4} + \frac{iw}{2} \ln\left(\frac{w}{2}\right)\right] \Gamma\left(1 - \frac{iw}{2}\right). \quad (12)$$

Every source derivative generated by D_β is another confluent hypergeometric function. The operator exponential can therefore be evaluated as an adaptively truncated series. Appendix C gives the derivation and the radial-derivative formula used in the calculation.

External convergence is included by an exact rescaling. Define

$$\lambda = 1 - \kappa, \quad \mathbf{y}' = \frac{\mathbf{y}}{\sqrt{\lambda}}, \quad \gamma' = \frac{\gamma}{\lambda}. \quad (13)$$

Changing variables to $\mathbf{x}' = \sqrt{\lambda} \mathbf{x}$ in Eq. (5) gives

$$F_{\kappa,\gamma}(w, \mathbf{y}; \beta) = \frac{1}{\lambda} \exp\left[\frac{iw}{2} \ln \lambda - \frac{iw\kappa}{2\lambda} |\mathbf{y}|^2\right] F_{0,\gamma'}(w, \mathbf{y}'; \beta). \quad (14)$$

The dependence on κ is therefore algebraic once the pure-shear series has been evaluated. No numerical contour integral appears in the final procedure.

III. STATIONARY PHASE AND THE ORIGIN OF THE FRINGES

The image positions satisfy

$$\nabla_{\mathbf{x}} \tau = 0, \quad \mathbf{y} = \mathbf{A} \mathbf{x} - \frac{\mathbf{x}}{|\mathbf{x}|^2}. \quad (15)$$

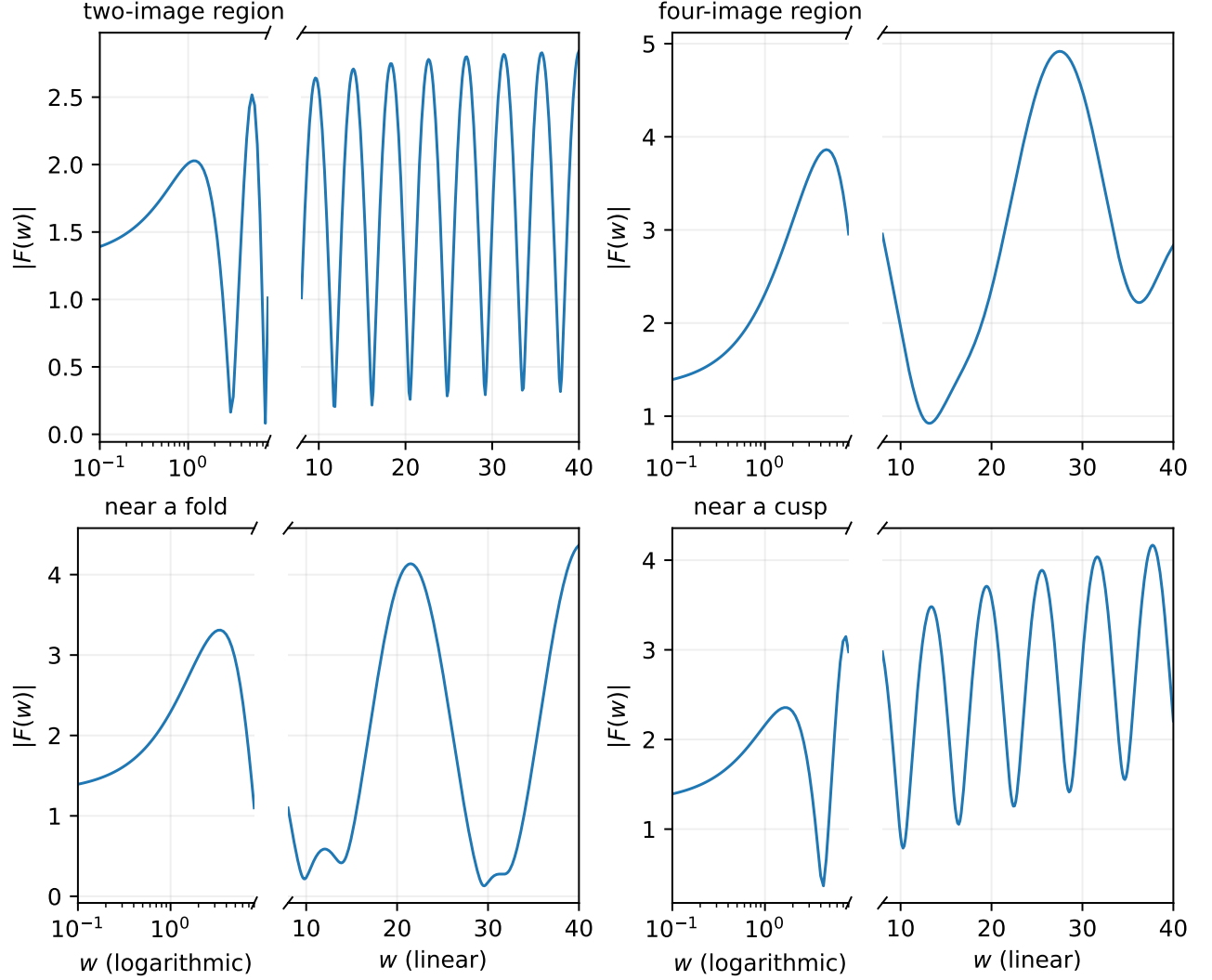


FIG. 1. Magnitude of the Chang–Refsdal amplification factor for representative source positions at $\kappa = \gamma = 0.2$ and $\beta = 0$. The low-frequency segment of each panel uses a logarithmic horizontal axis; the high-frequency segment uses a linear axis. The source positions are $\mathbf{y} = (0.55, 0)$, $(0.10, 0.10)$, $(-0.1389, 0.1793)$, and $(-0.395, 0)$ for the two-image, four-image, near-fold, and near-cusp panels, respectively. The curves are evaluated with the source-plane operator representation of Sec. II B.

Let $\mathbf{x}_j$ be a nondegenerate solution, $\tau_j = \tau(\mathbf{x}_j; \mathbf{y})$, and

$$\mathbf{H}_j = \nabla_{\mathbf{x}} \nabla_{\mathbf{x}} \tau|_{\mathbf{x}_j} \quad (16)$$

its Hessian. The signed magnification is $\mu_j = (\det \mathbf{H}_j)^{-1}$, and the Morse index n_j is the number of negative eigenvalues of $\mathbf{H}_j$. Applying stationary phase gives

$$F(w, \mathbf{y}) \sim \sum_j e^{i w \tau_j} H_j(w), \quad (17)$$

with

$$H_j(w) = \sqrt{|\mu_j|} e^{-i \pi n_j / 2} \left[1 + \frac{i C_{1j}}{w} + \frac{C_{2j}}{w^2} + O(w^{-3}) \right]. \quad (18)$$

The derivation and the explicit coefficients C_{1j} and C_{2j} are given in Appendix B. The higher-order terms improve

the resolved-image approximation after the asymptotic ordering has set in.

Figure 2 compares the operator result with the leading image sum and the expansion through w^{-2} in regular two-image and four-image regions. The four-image source was chosen so that none of the images is close to a fold. A separate example in Sec. V shows that the isolated-image expansion can remain poor at moderate frequency even when the source is formally inside a four-image region.

For two images, Eq. (17) gives

$$|F|^2 \simeq |H_1|^2 + |H_2|^2 + 2 \operatorname{Re} [H_1 H_2^* e^{i w (\tau_1 - \tau_2)}]. \quad (19)$$

If the kernels vary slowly over a fringe, the period is

$$\Delta w \simeq \frac{2\pi}{|\tau_1 - \tau_2|}. \quad (20)$$

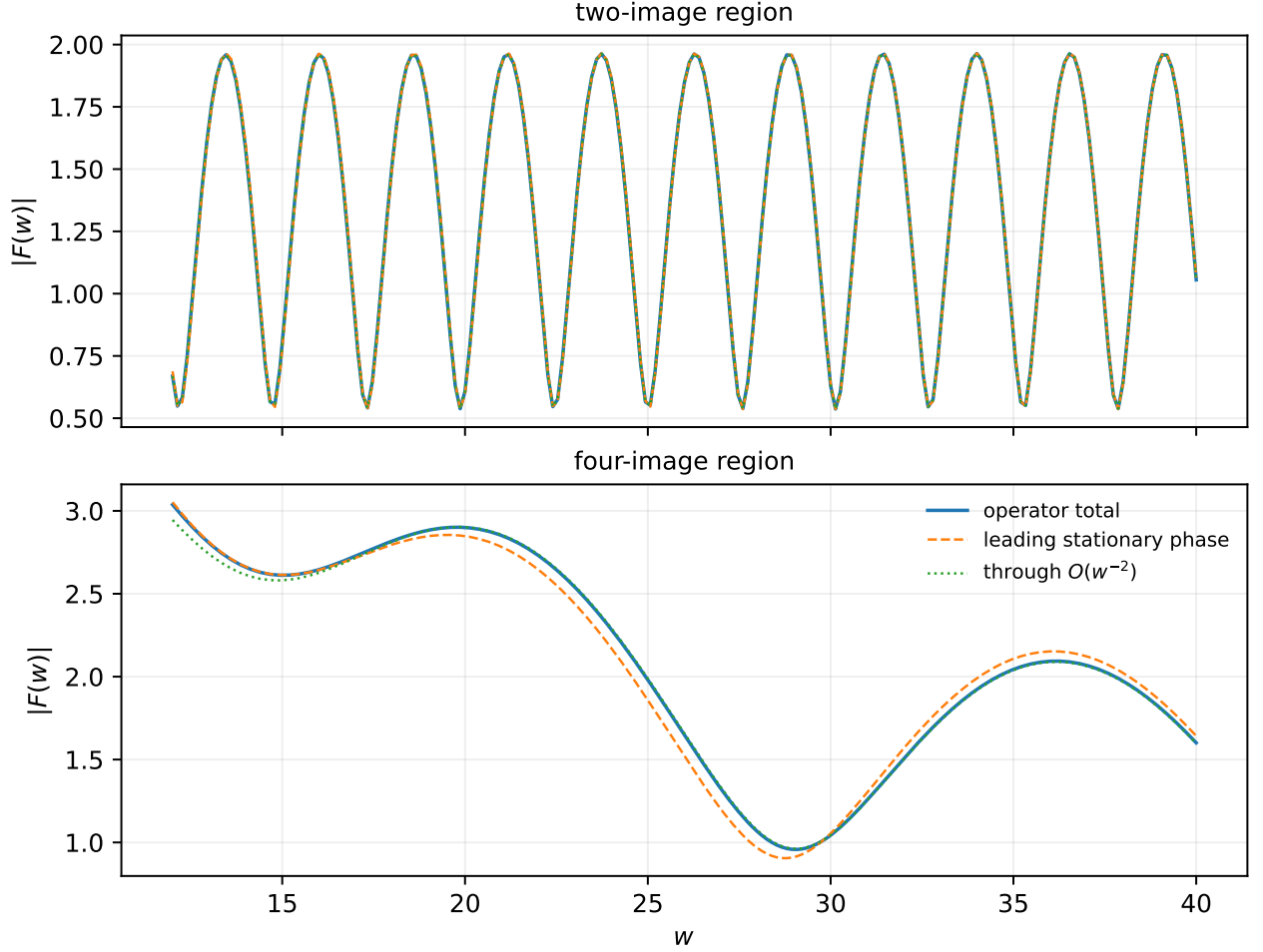


FIG. 2. Stationary-phase approximation in regular two-image and four-image regions for $\kappa = \gamma = 0.2$ and $\beta = 0$. The source positions are $\mathbf{y} = (0.90, 0)$ and $(0.03, 0.10)$, respectively. The plotted interval begins at $w = 12$, where the isolated-image expansion is asymptotically ordered for the selected sources. The w^{-2} result closely follows the operator total.

The same interpretation applies to every image pair in a multi-image region. The delay phases carry the rapid frequency dependence that must be separated before interpolation.

IV. WHY ORDINARY RELATIVE BINNING FAILS

For stationary Gaussian detector noise, define the complex overlap

$$\mathcal{Z}[a, b] = 4 \sum_f \frac{a(f)b^*(f)}{S_n(f)} \Delta f. \quad (21)$$

The log likelihood is

$$\ln \mathcal{L} = \text{Re } \mathcal{Z}[d, h] - \frac{1}{2} \mathcal{Z}[h, h] + \text{const}. \quad (22)$$

Relative binning chooses a fiducial waveform h_0 and writes

$$h(f; \vartheta) = r(f; \vartheta) h_0(f). \quad (23)$$

When r is smooth, it can be represented by a piecewise low-order polynomial, while the sums involving d and h_0 are precomputed [9].

For a lensed waveform,

$$r(f) = \frac{\tilde{h}_U(f; \lambda)}{\tilde{h}_U(f; \lambda_0)} \frac{F[w(f); \vartheta_L]}{F[w_0(f); \vartheta_{L,0}]}, \quad (24)$$

where λ denotes binary parameters and ϑ_L denotes lens parameters. The second factor need not be smooth. For two images,

$$F = e^{i w \tau_1} [H_1 + H_2 e^{i w \Delta \tau}], \quad \Delta \tau = \tau_2 - \tau_1, \quad (25)$$

so the lens ratio contains interference patterns with different delays in the candidate and fiducial models. Near a minimum of the fiducial amplification, the ratio can also acquire a sharp amplitude peak and a rapid phase excursion.

The loss of smoothness therefore occurs within a region of fixed image topology. It follows from the coherent addition of contributions with different parameter-dependent

delays, and does not require the creation or destruction of images. A numerical comparison of this direct ratio with the separated representation is given after the latter has been constructed in Sec. IX.

V. PROJECTED IMAGE CONTRIBUTIONS AT FIXED TOPOLOGY

The stationary-phase form suggests retaining the image delays as explicit phases. Within a region where the images can be labeled continuously, write

$$F(w; \vartheta_L) = \sum_{a=1}^{N_{\text{img}}} e^{i w \tau_a(\vartheta_L)} K_a(w; \vartheta_L). \quad (26)$$

The functions K_a will be called image contributions in this section. They approach the stationary-phase kernels H_a when the images are well resolved.

At finite frequency, using $K_a = H_a$ would impose the error of the stationary-phase expansion on the total. Instead, let $\hat{K}_a$ be any set of trial contributions and define

$$R(w) = F_{\text{op}}(w) - \sum_b e^{i w \tau_b} \hat{K}_b(w), \quad (27)$$

where F_{op} is the converged operator result. Choose smooth weights α_a with $\sum_a \alpha_a = 1$ and set

$$K_a(w) = \hat{K}_a(w) + \alpha_a e^{-i w \tau_a} R(w). \quad (28)$$

Substitution into Eq. (26) gives $F = F_{\text{op}}$ at every frequency. In a resolved region one may take $\hat{K}_a = H_a$. The residual then contains the finite-frequency part that is not described by the isolated saddles. Equation (28) is useful even when the image topology never changes.

Figure 3 uses a four-image source for which the isolated-image expansion is poorly ordered at moderate w . The projected sum agrees with the operator total to floating-point accuracy. The stationary-phase terms still determine the physical high-frequency limit, while the projection supplies the missing finite-frequency contribution.

Within a region of fixed topology, the candidate and fiducial waveforms can therefore be compared image by image. The difference between their image delays is retained analytically, while only the ratio of the projected amplitudes is interpolated. This construction addresses the interference fringes without relying on the accuracy of the isolated-image expansion at moderate frequency. Its remaining limitation is geometric: a physical image label cannot be followed through a caustic at which that image is created or destroyed.

VI. CHANGES IN IMAGE MULTIPLICITY

For the positive-parity Chang–Refsdal lens, a source outside the astroid caustic has two nondegenerate images

and a source inside has four [8]. Crossing a fold creates or destroys a pair. At a cusp, three images meet at a single critical point. The Hessian determinant of the participating images vanishes at the caustic, so the individual saddle kernels diverge even though the wave-optics amplification remains finite.

The left panel of Fig. 4 shows the fold and cusp trajectories used in the numerical tests. The signed path coordinate η is negative outside the caustic and positive inside for the conventions used here.

A fixed representation requires labels that exist before the real images do. For each unoccupied label, we use the nearest point on the critical curve as a geometric marker and assign the delay at that point. When images are created, their positions emerge from the marker and inherit its labels. The delay assigned to a label is kept at the critical-point delay while the cluster is unresolved. It approaches the physical image delay only after the delay separation can be resolved within the analysis band.

For a cluster with physical delays τ_j^{img} and critical delay τ_c , define the delay spread $\Delta\tau_{\text{cl}} = \max_j \tau_j^{\text{img}} - \min_j \tau_j^{\text{img}}$ and

$$B_c = S_c(w_{\text{max}} \Delta\tau_{\text{cl}}), \quad \tau_j^{\text{tr}} = (1 - B_c) \tau_c + B_c \tau_j^{\text{img}}. \quad (29)$$

Here S_c is a smooth transition function. Outside the caustic all labels in the cluster have $\tau_j^{\text{tr}} = \tau_c$. The right panels of Fig. 4 show the resulting delays for the selected paths. The tracked delays remain coincident through the unresolved neighborhood and then separate continuously toward the physical delays.

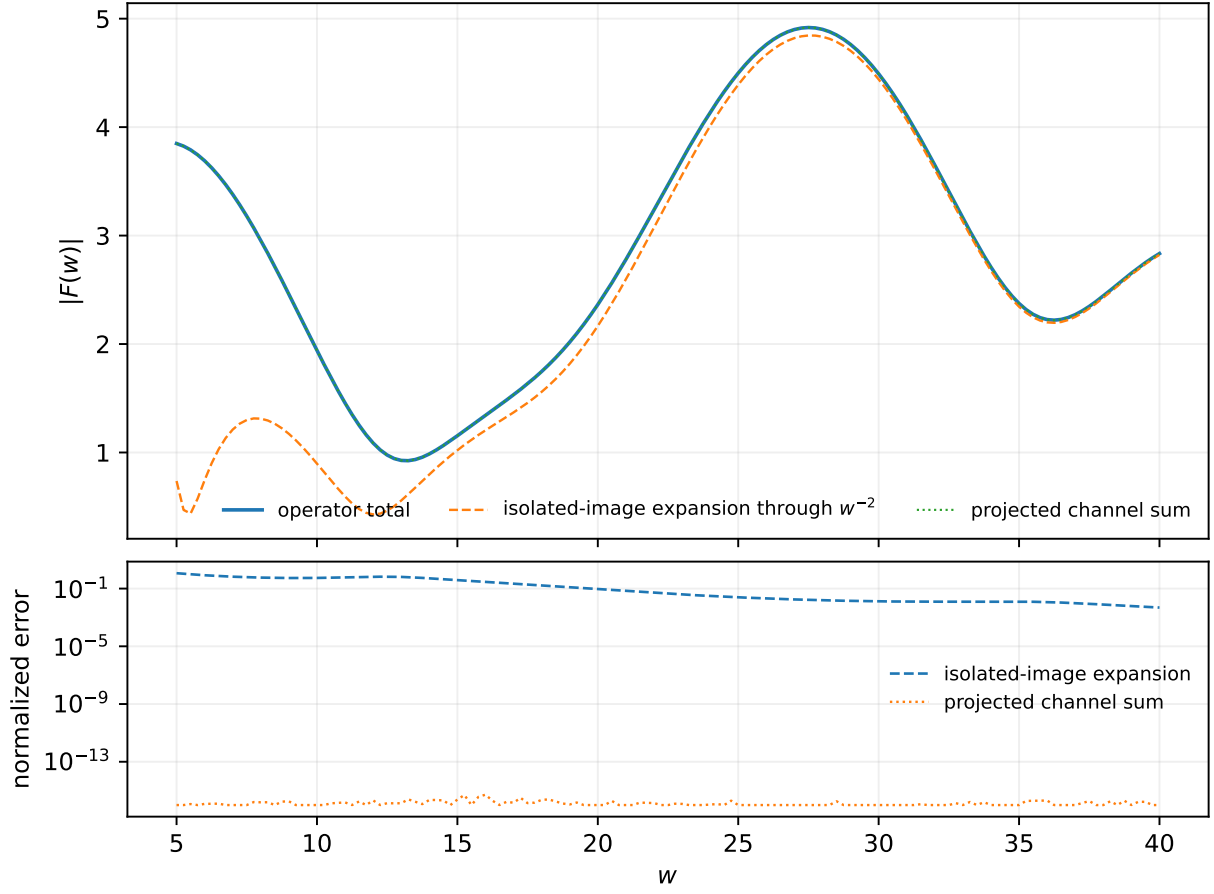


FIG. 3. A four-image example at $\mathbf{y} = (0.10, 0.10)$, $\kappa = \gamma = 0.2$. The isolated-image expansion through w^{-2} is inaccurate at moderate frequency because two of the saddles have small Hessian eigenvalues. The projected image sum uses the same saddle terms as targets and adds the residual in Eq. (28); its reconstruction error is below 9×10^{-16} in this calculation.

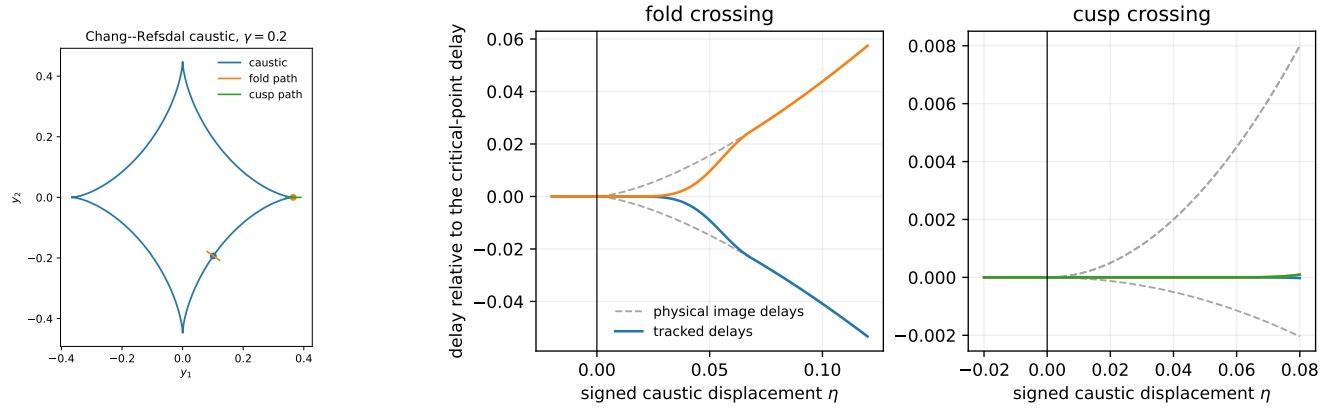


FIG. 4. Source trajectories and delay continuation across the Chang-Refsdal caustic. The left panel shows the astroid caustic at $\gamma = 0.2$ and the fold and cusp paths used in the numerical tests. The right panels show the physical image delays and the continuously tracked delays along those paths, measured relative to the critical-point delay. The vertical lines mark the caustic. The tracked delays remain coincident while the cluster is unresolved and approach the physical image delays farther inside the caustic.

VII. TOPOLOGY-STABLE FOUR-COMPONENT REPRESENTATION

The Chang–Refsdal lens has at most four nondegenerate images in the domain considered here. We retain four labels at every parameter point and continue them using their lens-plane positions. Labels without real images are placed at the critical point associated with the caustic cluster from which those images are created. The continuation procedure is described in Appendix D.

The exact projection is most useful when it is applied to the image cluster that is poorly described by isolated stationary points. Let C denote the active cluster and let P denote the remaining, persistent images. A generic fold has two labels in C , while a cusp has three. The persistent images are assigned their stationary-phase factors H_p . Their coherent sum is removed from the operator result, leaving the exact contribution associated with the cluster,

$$G_C(w) = F_{\text{op}}(w) - \sum_{p \in P} e^{i w \tau_p} H_p(w). \quad (30)$$

Any finite-frequency error in the persistent-image expansion is included in G_C by construction. This choice is preferable to distributing the same residual uniformly among all four labels, since it leaves the well-resolved image factors close to their physical stationary-phase limits.

For each cluster label $j \in C$, choose smooth weights α_j satisfying $\sum_{j \in C} \alpha_j = 1$. Equal weights are used in the calculations below. When the cluster is unresolved, define

$$L_j(w) = \alpha_j e^{-i w \tau_j} G_C(w). \quad (31)$$

The coherent sum of the L_j is exactly G_C for any assigned delays. If a cluster label corresponds to a real image, let H_j be its stationary-phase factor; for a virtual label set $H_j = 0$. For a real cluster image define

$$\delta_j = \min_{k \in C, k \neq j} |\tau_j - \tau_k| \quad (32)$$

and use the smooth switch

$$S_j(w) = \begin{cases} 0, & w\delta_j \leq \rho_0, \\ 10u_j^3 - 15u_j^4 + 6u_j^5, & \rho_0 < w\delta_j < \rho_1, \\ 1, & w\delta_j \geq \rho_1, \end{cases} \quad (33)$$

$$u_j = \frac{w\delta_j - \rho_0}{\rho_1 - \rho_0}. \quad (34)$$

For a virtual label, $S_j = 0$. We use $(\rho_0, \rho_1) = (0.5, 4)$ in the calculations below. The trial cluster factors are

$$\hat{K}_j = (1 - S_j)L_j + S_j H_j. \quad (35)$$

Their coherent sum need not equal G_C . The discrepancy is projected back onto the same cluster labels,

$$K_j = \hat{K}_j + \alpha_j e^{-i w \tau_j} \left[G_C - \sum_{k \in C} e^{i w \tau_k} \hat{K}_k \right], \quad j \in C. \quad (36)$$

For the persistent labels we set

$$K_p = H_p, \quad p \in P. \quad (37)$$

Equations (30)–(37) therefore give

$$F_{\text{op}}(w) = \sum_{a=1}^4 e^{i w \tau_a} K_a(w) \quad (38)$$

at every frequency.

The individual cluster factors can be written without the intermediate trial terms. Define

$$\bar{S}_C = \sum_{k \in C} \alpha_k S_k, \quad Q_C = \sum_{k \in C} S_k e^{i w \tau_k} H_k. \quad (39)$$

Then

$$K_j = S_j H_j + \alpha_j e^{-i w \tau_j} \times [(1 - S_j + \bar{S}_C)G_C - Q_C], \quad j \in C. \quad (40)$$

When the cluster is unresolved, $S_j = 0$ and the exact residual is divided among the coincident cluster labels. Once the images are resolved and the stationary-phase expansion becomes accurate, the correction in Eq. (36) decreases and K_j approaches H_j .

The labels and the active cluster are continued along a path in lens-parameter space. At a fold, the two cluster labels remain present on both sides of the caustic, although they correspond to real images only on the four-image side. Near a cusp, three labels are associated with the coalescing triplet. The construction retains a fixed total of four components while localizing the finite-frequency residual in the part of the image configuration for which the isolated-saddle description is least reliable.

External convergence requires no change to this projection. The operator total is obtained from Eq. (14); the delays, Hessians, and image positions are computed with the full matrix $\mathbf{A}$.

VIII. MULTI-COMPONENT RELATIVE BINNING

The global decomposition turns the lensed waveform into a sum of four fiducial components. Define

$$h_{a0}(f) = \tilde{h}_U(f; \lambda_0) e^{i w_0(f) \tau_{a0}} K_{a0}[w_0(f)]. \quad (41)$$

For a candidate lens and binary waveform,

$$\tilde{h}_L(f) = \sum_a r_a(f) h_{a0}(f), \quad (42)$$

where

$$r_a(f) = \frac{\tilde{h}_U(f; \lambda)}{\tilde{h}_U(f; \lambda_0)} \exp\{i[w(f)\tau_a - w_0(f)\tau_{a0}]\} \times \frac{K_a[w(f)]}{K_{a0}[w_0(f)]}. \quad (43)$$

This expression allows the point-mass redshifted mass to vary, in which case $w(f)$ differs between the candidate and fiducial models. The projected residual in Eq. (36) is already contained in K_a . There is no separate residual term in the likelihood.

Divide the frequency range into bins centered at f_b , and approximate each component ratio within a bin by

$$r_a(f) \simeq r_{a,b}^{(0)} + r_{a,b}^{(1)}(f - f_b). \quad (44)$$

The data-component summaries are

$$A_{a,b}^{(p)} = 4 \sum_{f \in b} \frac{d(f) h_{a0}^*(f)}{S_n(f)} (f - f_b)^p \Delta f, \quad p = 0, 1. \quad (45)$$

They give

$$\mathcal{Z}[d, h] \simeq \sum_{a,b} \left[A_{a,b}^{(0)} r_{a,b}^{(0)*} + A_{a,b}^{(1)} r_{a,b}^{(1)*} \right]. \quad (46)$$

The waveform norm contains pairs of components. Define

$$B_{ac,b}^{(p)} = 4 \sum_{f \in b} \frac{h_{a0}(f) h_{c0}^*(f)}{S_n(f)} (f - f_b)^p \Delta f, \quad p = 0, 1, 2. \quad (47)$$

Multiplying the two linear expansions gives

$$\begin{aligned} \mathcal{Z}[h, h] \simeq \sum_{a,c,b} & \left\{ B_{ac,b}^{(0)} r_{a,b}^{(0)*} r_{c,b}^{(0)*} \right. \\ & + B_{ac,b}^{(1)} \left[r_{a,b}^{(1)} r_{c,b}^{(0)*} + r_{a,b}^{(0)} r_{c,b}^{(1)*} \right] \\ & \left. + B_{ac,b}^{(2)} r_{a,b}^{(1)} r_{c,b}^{(1)*} \right\}. \end{aligned} \quad (48)$$

The Hermitian relation $B_{ca,b}^{(p)} = B_{ac,b}^{(p)*}$ reduces the number of independent component pairs. For four labels, ten pair summaries are required per moment. The factor-of-several increase in precomputed summaries is small compared with the reduction in frequency nodes when the complete amplification ratio is strongly oscillatory.

Equation (43) separates two kinds of frequency dependence. The ratio K_a/K_{a0} contains the finite-frequency response, including the projected residual. The exponential contains the known difference between the candidate and fiducial image delays. We retain this exponential analytically and interpolate only the unlensed-waveform ratio and K_a/K_{a0} . Since w is proportional to frequency, write

$$w(f)\tau_a - w_0(f)\tau_{a0} = 2\pi f \delta t_a \quad (49)$$

with a frequency-independent relative delay δt_a , and define

$$q_a(f) = \frac{\tilde{h}_U(f; \lambda)}{\tilde{h}_U(f; \lambda_0)} \frac{K_a[w(f)]}{K_{a0}[w_0(f)]}. \quad (50)$$

Then $r_a = e^{2\pi i f \delta t_a} q_a$. Approximating only q_a within each bin replaces the summaries in Eqs. (45) and (47) by the same sums evaluated at the time shifts δt_a and $\delta t_a - \delta t_c$, respectively. These shifted sums are short discrete Fourier transforms and can be tabulated on a delay grid or evaluated with FFTs. This phase-aware form is useful when the candidate and fiducial delays differ enough for the exponential to accumulate several cycles across the detector band. The numerical benchmarks below retain this delay phase exactly and interpolate only q_a .

IX. NUMERICAL RESULTS

The tests use $\gamma = 0.2$ and $5 \leq w \leq 40$ unless stated otherwise. The operator series is truncated after four successive terms fall below 2×10^{-12} times the partial sum, with a maximum order chosen high enough that every evaluation in the reported local tests satisfies this criterion. The image geometry is obtained from the quartic reduction in Appendix A. The numerical interpolation diagnostic is

$$\epsilon(w) = \frac{|F_{\text{int}}(w) - F_{\text{op}}(w)|}{\max[|F_{\text{op}}(w)|, 0.15 \max_{w'} |F_{\text{op}}(w')|]}. \quad (51)$$

The floor prevents a zero of the exact amplification from dominating the reported maximum. It does not enter the channel construction.

A. Fixed-topology comparison

Figure 5 compares the two representations for sources that remain outside the same generic fold. Both models have two real images, yet the ratio of their complete amplification factors develops a sequence of narrow peaks and phase excursions because the interference minima occur at different frequencies. Once the four contributions are separated, the ratios of their complex amplitudes vary slowly across the same interval. The known delay differences are treated analytically and are therefore absent from the lower panels.

A greedy interpolation test starts with the two endpoints and repeatedly adds the frequency with the largest value of Eq. (51). The differences between the candidate and fiducial delays are retained exactly; only the slowly varying factor ratios shown in the lower panels of Fig. 5 are interpolated. The separated representation reaches $\max_w \epsilon < 10^{-3}$ with 11 frequency nodes. Direct interpolation of F/F_0 requires 135 of the 141 validation frequencies for the same criterion.

B. Fold and cusp crossings

For the local crossing tests, two persistent contributions and a two-term cluster are used at a fold; one persistent

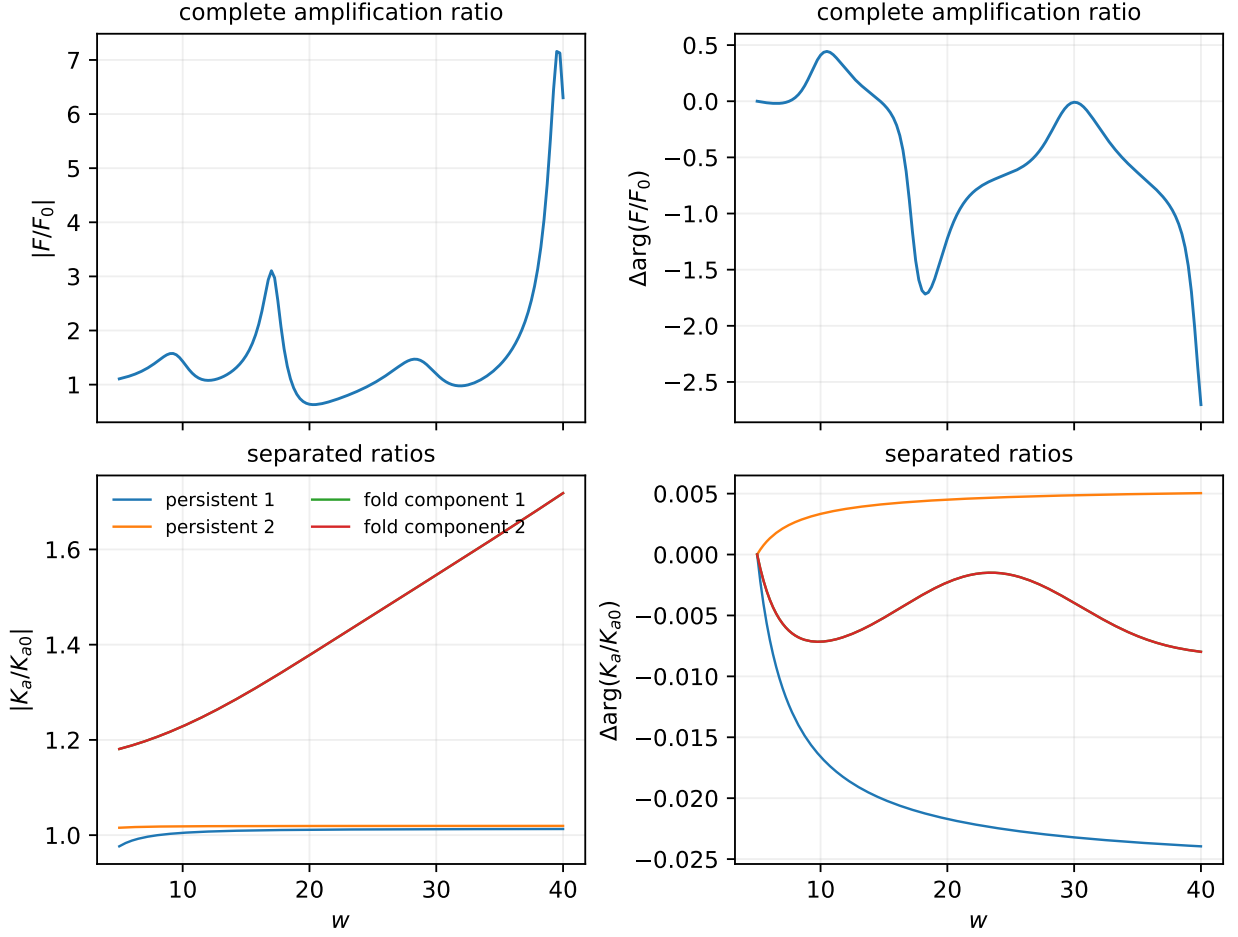


FIG. 5. Direct and separated candidate-to-fiducial ratios within a fixed two-image topology. The two sources lie outside the same generic fold at signed normal displacements $\eta_s = -0.040$ and $\eta_s = -0.010$, with $\gamma = 0.2$. The complete amplification ratio in the upper panels develops a sequence of sharp peaks and phase excursions as the interference minima shift. The lower panels show the corresponding ratios of the two persistent image amplitudes and the two virtual fold-cluster amplitudes. Their phases are measured relative to the lowest plotted frequency. The known differences between the candidate and fiducial delays are omitted from the lower panels and retained analytically when the four contributions are recombined.

contribution and a three-term cluster are used at a cusp. These use the cluster-local projection in Eqs. (30)–(36). Figure 6 shows the candidate-to-fiducial ratios for complete outside-to-inside crossings. Several cluster curves coincide because equal weights are used while the corresponding delays are unresolved.

Table I gives the number of adaptively selected frequency nodes needed to reach $\max_w \epsilon < 10^{-3}$. The separated representation needs 7–10 nodes for the four representative crossings. Direct interpolation of the complete ratio requires 70 or 71 of the 71 validation frequencies. Across all crossing pairs in the scan $-0.02 \leq \eta \leq 0.02$, the separated representation uses 6–10 nodes.

Figure 7 shows the error as the node budget is increased. The direct-ratio curves remain above the target on the displayed node scale because they must resolve the moving fringes.

The coherent component sum reproduces the operator

TABLE I. Frequency nodes required for a maximum normalized interpolation error of 10^{-3} over $5 \leq w \leq 40$.

Crossing	separated ratios	complete ratio
Fold, outside to inside	9	70
Fold, inside to outside	10	70
Cusp, outside to inside	7	71
Cusp, inside to outside	7	71

total to between 4.4×10^{-16} and 6.7×10^{-16} in the four local crossing calculations. The smallest component magnitudes remain above 0.47 in the cusp tests and 0.59 in the fold tests, so the candidate-to-fiducial ratios are not made ill conditioned by a vanishing denominator.

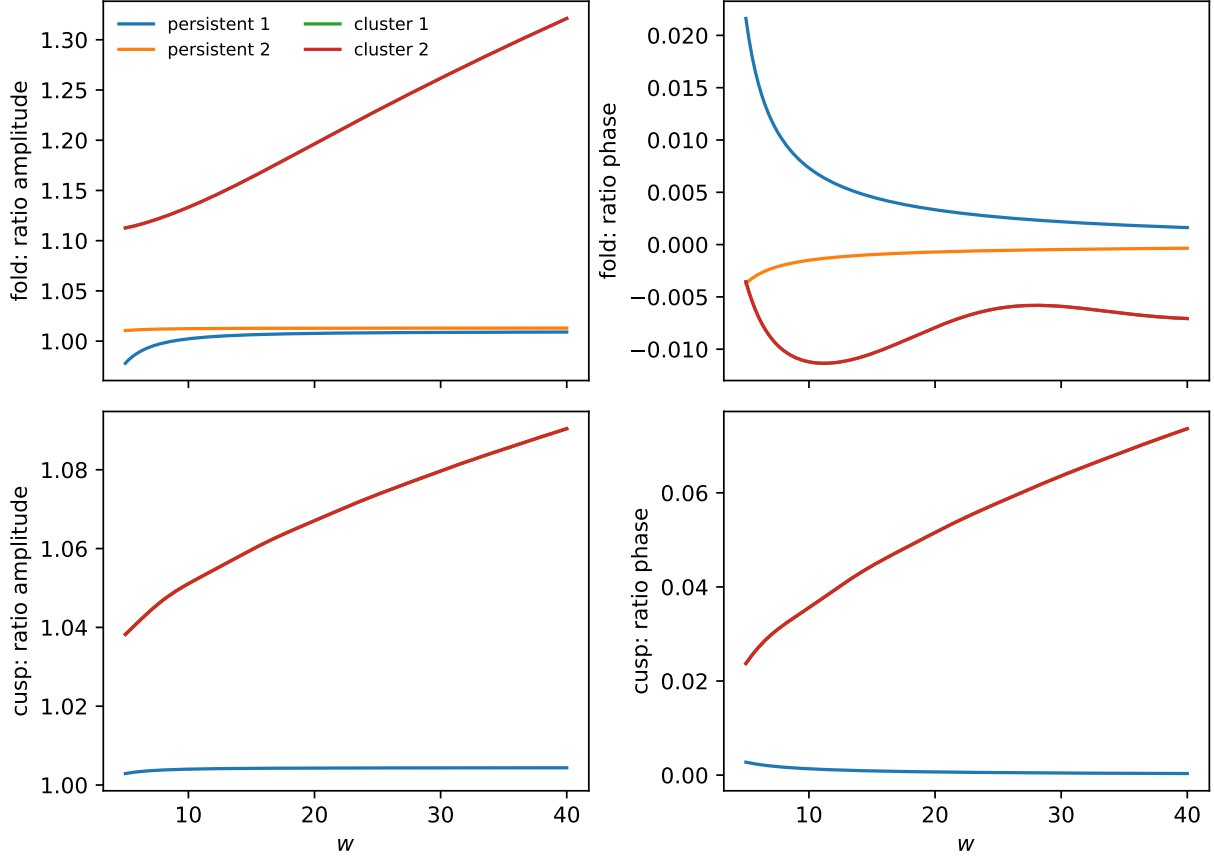


FIG. 6. Amplitude and unwrapped phase of the separated ratios for a fold crossing and a cusp crossing from $\eta = -0.01$ to $\eta = +0.01$. The ratios remain smooth even though the number of real images changes.

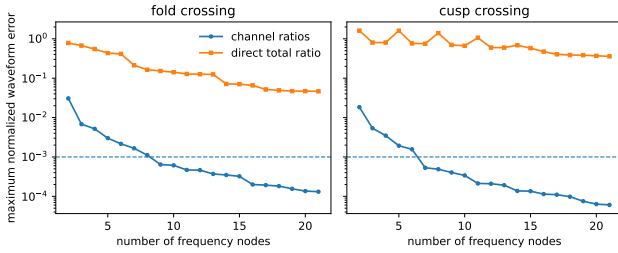


FIG. 7. Maximum interpolation error versus frequency-node budget for a full fold crossing and a full cusp crossing. The dashed line marks the 10^{-3} target.

C. Geometry, convergence, and global tracking

The quartic image solver was compared with a deterministic two-dimensional multistart solver on 168 configurations, including general source-shear orientations and points within 10^{-5} of selected folds and cusps. The image count agreed in every case. The maximum matched position difference was 2.1×10^{-12} , and the maximum lens-equation residual of a quartic image was 1.8×10^{-13} .

In this scan the quartic calculation was about 190 times faster than the multistart calculation. The timing indicates that image finding is negligible compared with the special-function evaluation in the current prototype.

The external-convergence mapping was tested for six values between $\kappa = 0.2$ and 0.3 , while keeping the reduced shear and scaled source position fixed. The four-term reconstruction errors ranged from 2.0×10^{-15} to 6.3×10^{-15} .

A separate continuation path varied the source position, shear magnitude, and shear orientation through two-image and four-image regions. The path contains four changes in image count. All four labels remained defined. The largest absolute reconstruction error over the 49 path points was 9.4×10^{-11} ; this occurred where large intermediate terms produced strong floating-point cancellation. Away from those points the errors were near 10^{-15} . The continuation algorithm is designed for nearby parameter points. A distant independent proposal should be initialized from a fixed convention or connected to the fiducial point by a short path.

X. DISCUSSION

The failure of ordinary relative binning in wave-optics lensing is caused by the coexistence of several parameter-dependent image phases. It does not require a caustic crossing. Separating those phases produces functions whose candidate-to-fiducial ratios are much smoother. The residual projection in Eq. (28) makes this statement useful at finite frequency, where the isolated-image expansion may be inaccurate. The stationary-phase approximation supplies the physical resolved-image limit, while the operator result fixes the coherent sum at every frequency.

The topology-stable extension addresses a distinct issue. Physical images cannot be labeled globally because pairs are created at folds and triplets meet at cusps. Four continuously tracked labels are sufficient for a positive-parity Chang–Refsdal lens. Virtual labels are placed at a critical point, and the representation remains finite while their physical images are absent. Once the images are resolved, the corresponding terms approach the usual geometrical contributions.

The numerical tests are intended to establish the representation and its interpolation behavior over a controlled domain. They do not yet constitute a full parameter-estimation study. The source positions, convergence, and shear were varied, while the interpolation tables in Table I keep the redshifted microlens mass fixed. Equation (43) includes mass variation and ordinary binary parameters, so these can be incorporated in the same relative-binning summaries. Their joint effect on bin placement should be tested with detector noise curves and complete compact-binary waveform models.

The present operator implementation is restricted to positive-parity macro images with $1 - \kappa > |\gamma|$. Macro saddles are important for strongly lensed events and require a separate treatment. The source-derivative series also becomes more demanding as the reduced shear approaches unity or as the frequency is extended far beyond the range studied here. Stable recurrences or a reduced representation of the operator total would be useful in that regime.

The decomposition is not specific to relative binning. It provides a set of frequency functions with explicit rapidly varying phases and smooth parameter dependence. The same representation may be useful for surrogate models, reduced-order quadrature, and interpolation of microlensing amplification factors. For the application considered here, its immediate consequence is a multi-component relative-binning likelihood that remains well defined across the Chang–Refsdal caustic network and contains no numerical lens-plane integral.

XI. CONCLUSION

We have constructed a decomposition of the positive-parity Chang–Refsdal amplification factor in which the image-delay phases are retained explicitly. Within a re-

gion of fixed topology, an exact residual projection supplements the isolated-image stationary-phase terms and reproduces the complete wave-optics result. Four continuously tracked labels extend the decomposition through folds and cusps. External convergence is included by an exact mass-sheet rescaling, and the all-frequency amplification is evaluated from the point-mass solution through a source-derivative expansion.

The separated candidate-to-fiducial ratios are smooth in the fixed-topology and caustic-crossing examples studied here. Complete fold and cusp crossings reach a maximum interpolation error of 10^{-3} with 6–10 frequency nodes, whereas direct interpolation of the total amplification ratio requires nearly the full validation grid. The likelihood formulas in Eqs. (46) and (48) show how these ratios enter relative binning without a separate treatment of the projected finite-frequency residual.

DATA AND CODE AVAILABILITY

The scripts, numerical data, and contour-free implementation used to produce the figures and tables accompany this manuscript. The numerical lens-plane integral is not used by the code path described in the paper.

Appendix A: Quartic image geometry

The image equation can be reduced to a quartic polynomial. This appendix gives the reduction for a general symmetric matrix $\mathbf{A}$, so convergence and an arbitrarily oriented shear are included.

Let $Y = |\mathbf{y}|$ and rotate to the source-aligned basis $\mathbf{B} = (\hat{\mathbf{y}}, \hat{\mathbf{y}}_\perp)$. In this basis,

$$\mathbf{y}' = (Y, 0), \quad \mathbf{A}' = \mathbf{B}^\top \mathbf{A} \mathbf{B} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}. \quad (\text{A1})$$

Set $u = 1/|\mathbf{x}'|^2$. The lens equation becomes

$$(\mathbf{A}' - u\mathbf{1})\mathbf{x}' = (Y, 0). \quad (\text{A2})$$

With

$$D(u) = (a - u)(d - u) - b^2, \quad (\text{A3})$$

the generic solution for a fixed u is

$$x'_1 = \frac{Y(d - u)}{D(u)}, \quad x'_2 = -\frac{Yb}{D(u)}. \quad (\text{A4})$$

Imposing $|\mathbf{x}'|^2 = u^{-1}$ gives

$$D(u)^2 - Y^2 u [(d - u)^2 + b^2] = 0. \quad (\text{A5})$$

Expanding,

$$\begin{aligned} 0 = & u^4 + [-2(a + d) - Y^2]u^3 \\ & + [a^2 + 4ad + d^2 - 2b^2 + 2dY^2]u^2 \\ & + [-2(a + d)(ad - b^2) - Y^2(d^2 + b^2)]u + (ad - b^2)^2. \end{aligned} \quad (\text{A6})$$

Positive real roots are reconstructed with Eq. (A4), rotated back to the original frame, and polished with deterministic Newton iterations. The polishing corrects floating-point error in the algebraic roots; it is not an image search.

When $b = 0$, Eq. (A4) has a removable singularity for the symmetric off-axis pair. The axial images solve

$$ax_1'^2 - Yx_1' - 1 = 0, \quad (\text{A7})$$

and an off-axis pair, when present, satisfies

$$u = d, \quad x_1' = \frac{Y}{a-d}, \quad x_2'^2 = \frac{1}{d} - x_1'^2. \quad (\text{A8})$$

At $Y = 0$ and nonzero shear, the images lie along the eigenvectors of $\mathbf{A}$ at radii equal to the inverse square roots of its eigenvalues. The exactly unsheared and aligned case is an Einstein ring and is treated as a degeneracy.

Appendix B: Stationary-phase coefficients

Write $\mathbf{x} = \mathbf{x}_j + \boldsymbol{\xi}$ near a nondegenerate image and expand

$$\tau = \tau_j + \frac{1}{2}H_{ab}\xi_a\xi_b + \frac{1}{3!}T_{abc}\xi_a\xi_b\xi_c + \frac{1}{4!}Q_{abcd}\xi_a\xi_b\xi_c\xi_d + \dots \quad (\text{B1})$$

After the scaling $\boldsymbol{\xi} = w^{-1/2}\mathbf{z}$, the quadratic term is independent of w and the higher derivatives generate an

expansion in powers of w^{-1} . The oscillatory Gaussian moments are evaluated with covariance $i\mathbf{H}^{-1}$. At first order, Wick contractions give

$$C_1 = -\frac{1}{8}Q_{abcd}G_{ab}G_{cd} + \frac{1}{12}T_{abc}T_{def}G_{ad}G_{be}G_{cf} + \frac{1}{8}T_{abc}T_{def}G_{ab}G_{de}G_{cf}, \quad G = \mathbf{H}^{-1}. \quad (\text{B2})$$

The second coefficient receives terms from derivatives through sixth order and from products of the lower-order vertices. For the Chang–Refsdal potential, all third and higher derivatives arise from $-\ln|\mathbf{x}|$. This allows the contractions to be reduced to a polynomial in three scaled inverse-Hessian entries.

Let $\mathbf{e}_r = \mathbf{x}_j/|\mathbf{x}_j|$ and let $\mathbf{e}_t$ be the positively oriented orthogonal vector. Define

$$\mathbf{M} = \begin{pmatrix} \mathbf{e}_r^\top \\ \mathbf{e}_t^\top \end{pmatrix} \mathbf{H}_j^{-1} \begin{pmatrix} \mathbf{e}_r & \mathbf{e}_t \end{pmatrix}, \quad (\text{B3})$$

$$u = |\mathbf{x}_j|^{-2}, \quad p = uM_{rr}, \quad q = uM_{rt}, \quad s = uM_{tt}. \quad (\text{B4})$$

Then

$$C_1 = \frac{1}{12}(10p^3 - 12p^2s - 9p^2 - 48pq^2 + 18ps^2 + 18ps + 72q^2s + 36q^2 - 9s^2). \quad (\text{B5})$$

The second coefficient is $C_2 = -P_2/288$, with the polynomial below.

$$\begin{aligned} P_2 = & 1540p^6 - 1680p^5s - 3780p^5 - 16800p^4q^2 + 2520p^4s^2 + 5040p^4s + 2961p^4 \\ & + 40320p^3q^2s + 40320p^3q^2 - 3600p^3s^3 - 8280p^3s^2 - 5364p^3s - 720p^3 \\ & + 40320p^2q^4 - 64800p^2q^2s^2 - 99360p^2q^2s - 32184p^2q^2 + 3780p^2s^4 + 10800p^2s^3 \\ & + 9126p^2s^2 + 2160p^2s - 86400pq^4s - 66240pq^4 + 60480pq^2s^3 + 129600pq^2s^2 \\ & + 73008pq^2s + 8640pq^2 - 3780ps^4 - 5940ps^3 - 2160ps^2 - 11520q^6 + 60480q^4s^2 \\ & + 86400q^4s + 24336q^4 - 30240q^2s^3 - 35640q^2s^2 - 8640q^2s + 945s^4 + 720s^3. \end{aligned} \quad (\text{B6})$$

Substitution of Eqs. (B5) and (B6) into Eq. (18) gives the resolved-image targets used in the main text.

Appendix C: Source-plane operator representation

For $\kappa = 0$, write the shear contribution to the Fermat potential as $-(\gamma/2)x_iQ_{ij}x_j$, where Q is the traceless matrix in Eq. (3) divided by γ . After removing the source phase, the remaining lens-plane integral contains $\exp(-i w \mathbf{y} \cdot \mathbf{x})$. Acting on this Fourier kernel,

$$\partial_{y_i}\partial_{y_j}e^{-i w \mathbf{y} \cdot \mathbf{x}} = -w^2x_ix_je^{-i w \mathbf{y} \cdot \mathbf{x}}. \quad (\text{C1})$$

Multiplication by the quadratic shear phase is therefore equivalent to the source-space operator in Eq. (9). Expanding its exponential gives

$$\mathcal{G}_\gamma = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\gamma}{2w} \right)^n D_{\beta}^n \mathcal{G}_{\text{PM}}. \quad (\text{C2})$$

The point-mass seed depends only on $s = |\mathbf{y}|^2$. Its

radial derivatives are

$$\frac{d^k \mathcal{G}_{\text{PM}}}{ds^k} = C(w) \left(-\frac{iw}{2} \right)^k \frac{(1 - iw/2)_k}{k!} \times {}_1F_1 \left(1 - \frac{iw}{2} + k; 1 + k; -\frac{iws}{2} \right). \quad (\text{C3})$$

Repeated application of D_β is represented as a finite sum of monomials in $z = y_1 + iy_2$, $\bar{z}$, and the derivatives in Eq. (C3). The series is stopped after a prescribed number of consecutive terms fall below the requested fraction of the current partial sum. The order used and the last-term estimate are retained as convergence diagnostics.

For completeness, the convergence rescaling follows by substituting $\mathbf{x}' = \sqrt{\lambda} \mathbf{x}$ into Eq. (1):

$$\tau_{\kappa, \gamma}(\mathbf{x}; \mathbf{y}) = \tau_{0, \gamma/\lambda} \left(\mathbf{x}'; \frac{\mathbf{y}}{\sqrt{\lambda}} \right) + \frac{1}{2} \ln \lambda - \frac{\kappa}{2\lambda} |\mathbf{y}|^2. \quad (\text{C4})$$

The Jacobian $d^2x = \lambda^{-1} d^2x'$ then gives Eq. (14).

Appendix D: Global label continuation

At a given source position, the real images are found from Eq. (A6). The nearest caustic point is obtained by a one-dimensional minimization along the critical curve. Unoccupied labels are placed at the corresponding critical point in the lens plane.

Suppose the previous parameter point has marker positions $\mathbf{m}_a$, one for each of the four labels, and the current point has N real images $\mathbf{x}_j$. The assignment is chosen by minimizing

$$\sum_{j=1}^N \frac{|\mathbf{x}_j - \mathbf{m}_{\pi(j)}|^2}{\ell^2} \quad (\text{D1})$$

over injections π from the images to the four labels. The scale ℓ is taken from the median marker radius and only makes the cost dimensionless. Since there are at most four images, every assignment can be checked directly. The markers of occupied labels are updated to the image positions; the others are set to the nearest critical point.

At the first point on a path, the real images are ordered by polar angle and then by radius. This convention is sufficient to initialize a continuous trajectory. It does not define a globally path-independent permutation around all possible loops in parameter space. For likelihood proposals that are far from the fiducial point, one may reset the labels with the initialization rule or evaluate the model along a short interpolation path from the fiducial parameters. The coherent sum in Eq. (38) is independent of the label permutation; only the smoothness of individual ratios depends on consistent continuation.

-
- [1] T. T. Nakamura and S. Deguchi, *Progress of Theoretical Physics Supplement* **133**, 137 (1999).
 - [2] R. Takahashi and T. Nakamura, *The Astrophysical Journal* **595**, 1039 (2003), [arXiv:astro-ph/0305055](#).
 - [3] P. Christian, S. Vitale, and A. Loeb, *Physical Review D* **98**, 103022 (2018), [arXiv:1802.02586 \[astro-ph.HE\]](#).
 - [4] A. K. Meena and J. S. Bagla, *Monthly Notices of the Royal Astronomical Society* **492**, 1127 (2020), [arXiv:1903.11809 \[astro-ph.CO\]](#).
 - [5] A. Mishra, A. K. Meena, A. More, S. Bose, and J. S. Bagla, *Monthly Notices of the Royal Astronomical Society* **508**, 4869 (2021), [arXiv:2102.03946 \[astro-ph.CO\]](#).
 - [6] K. Chang and S. Refsdal, *Nature* **282**, 561 (1979).
 - [7] K. Chang and S. Refsdal, *Astronomy and Astrophysics* **132**, 168 (1984).
 - [8] J. H. An and N. W. Evans, *Monthly Notices of the Royal Astronomical Society* **369**, 317 (2006), [arXiv:astro-ph/0601457](#).
 - [9] B. Zackay, L. Dai, and T. Venumadhav, *arXiv e-prints* (2018), [arXiv:1806.08792 \[astro-ph.IM\]](#).
 - [10] L. Dai, T. Venumadhav, and B. Zackay, *arXiv e-prints* (2018), [arXiv:1806.08793 \[gr-qc\]](#).
 - [11] K. Krishna, A. Vijaykumar, A. Ganguly, C. Talbot, S. Biscoveanu, R. N. George, N. Williams, and A. Zimmerman, *arXiv e-prints* (2023), [arXiv:2312.06009 \[gr-qc\]](#).
 - [12] H. Narola, J. Janquart, L. Haegel, K. Haris, O. A. Hannuksela, and C. Van Den Broeck, *Physical Review D* **110**, 084085 (2024).
 - [13] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, Berlin, 1992).